Structural-Graphical Causal Inference

(SICSS UCLA — Spring 2026)

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All You Need is *Data*? Simpson's Paradox

	HbA1c low (Y)	HbA1c high ($\neg Y$)		Sucess Rate
drug (X)	20	20	40	50%
no-drug ($\neg X$)	16	24	40	40%
	36	44		

$$P(Y \mid F, X) < P(Y \mid F, \neg X)$$

$$P(Y \mid \neg F, X) < P(Y \mid \neg F, \neg X)$$

but

$$P(Y \mid X) > P(Y \mid \neg X) \,!$$

male ($\neg F$)	HbA1c low (Y)	HbA1c high ($\neg Y$)		Sucess Rate
drug (X)	18	12	30	60%
no-drug ($\neg X$)	7	3	10	70%
	25	15	40	

female (F)	HbA1c low (Y)	HbA1c high ($\neg Y$)		Sucess Rate
drug (X)	2	8	10	20%
no-drug ($\neg X$)	9	21	30	30%
	11	29	40	

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weight loss ($\neg F$)	HbA1c low (Y)	HbA1c high ($\neg Y$)		Success Rate
drug (X)	18	12	30	60%
no-drug ($\neg X$)	7	3	10	70%
	25	15	40	

no weight loss (F)	HbA1c low (Y)	HbA1c high ($\neg Y$)		Success Rate
drug (X)	2	8	10	20%
no-drug ($\neg X$)	9	21	30	30%
	11	29	40	

Are the result different with a different Story?

$(\neg F)$	HbA1c low (Y)	HbA1c high $(\neg Y)$		Sucess Rate
drug (X)	18	12	30	60%
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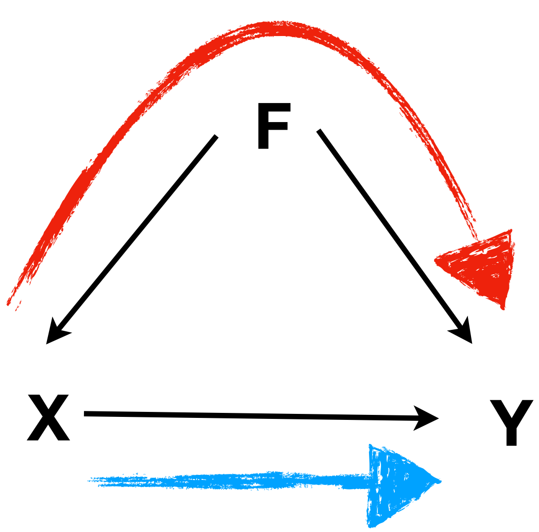
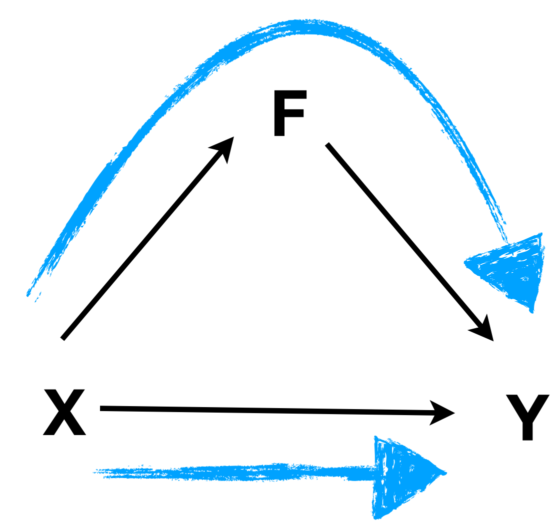
Sex-Stratified



Weight Loss-Stratified



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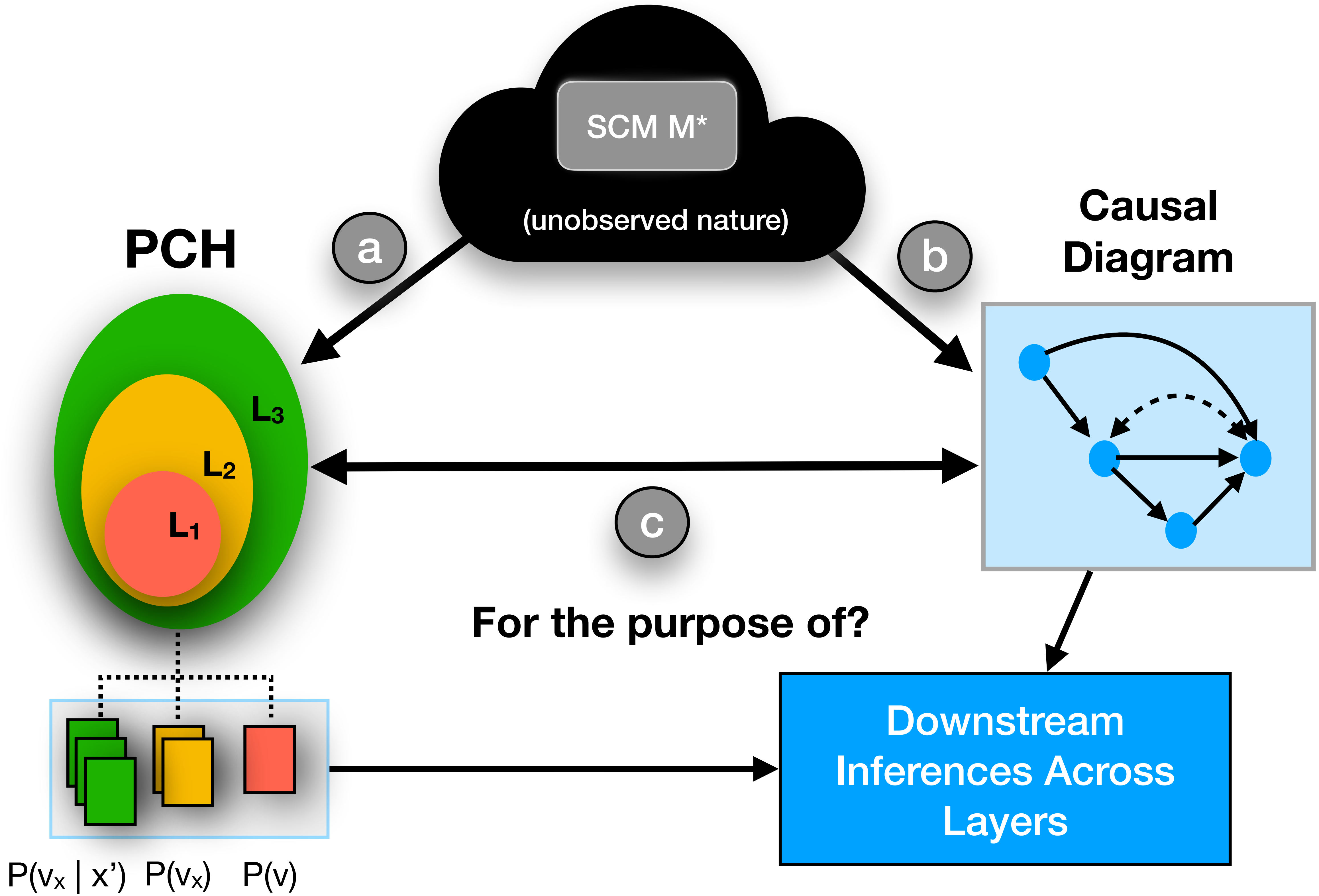
Weight Loss-Stratified

Data alone is insufficient for causal inference.

Structural-Graphical Approach: Big Picture



Structural Causal Models

Definition: A structural causal model (SCM) M is a 4-tuple $<\boldsymbol{V}, \boldsymbol{U}, \mathscr{F}, P(\boldsymbol{u})>$, where

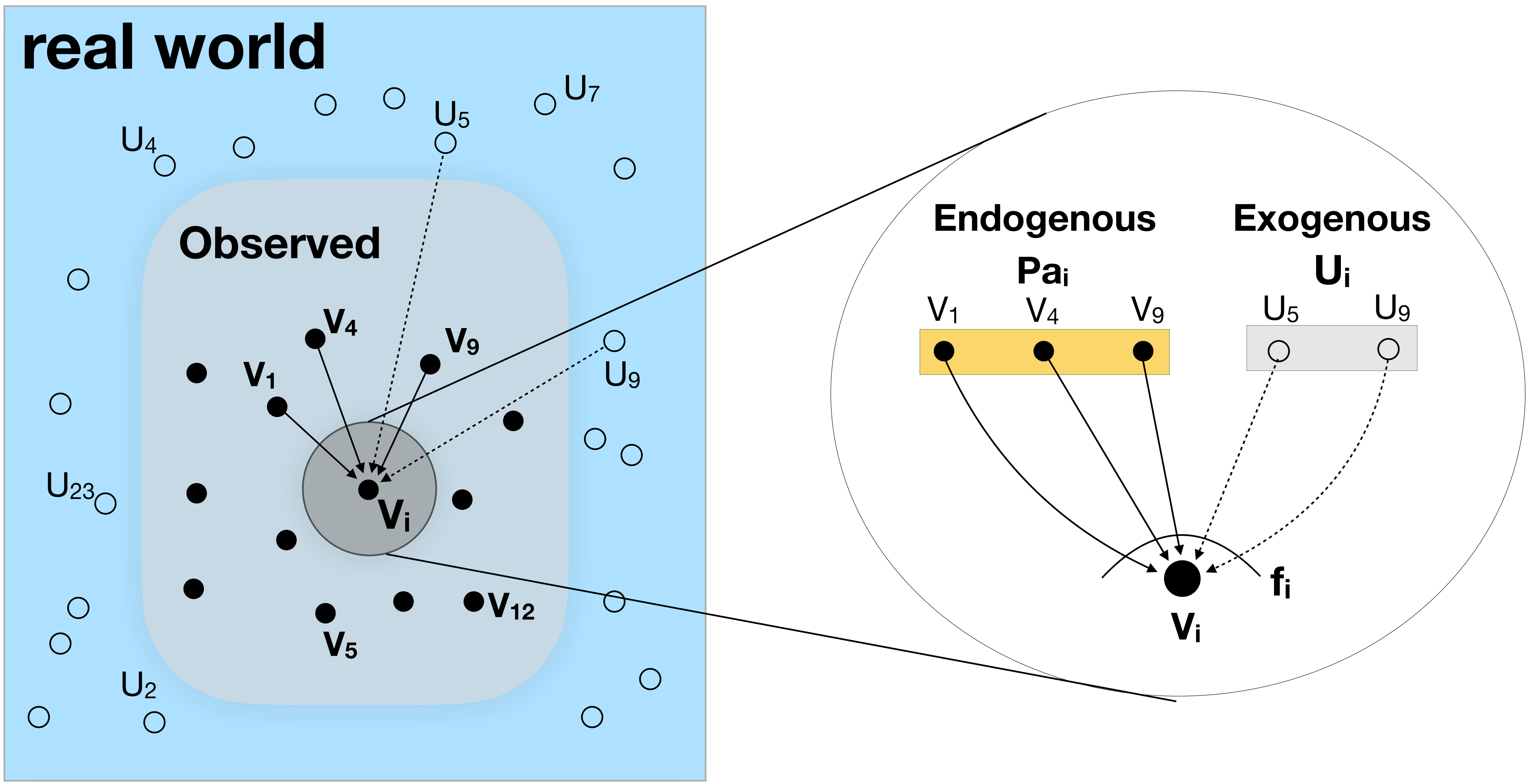
- $\boldsymbol{V} = \{V_1,\ldots,V_n\}$ are endogenous variables;
- $\boldsymbol{U} = \{U_1,\ldots,U_m\}$ are exogenous variables;
- $\mathscr{F} = \{f_1,\ldots, f_n\}$ are functions determining $\boldsymbol{V}$,
  $\boxed{v_i \leftarrow f_i(pa_i, u_i),}\ Pa_i \subset V_i, U_i \subset U;$
- $P(\boldsymbol{u})$ is a distribution over $\boldsymbol{U}$

Not regression!

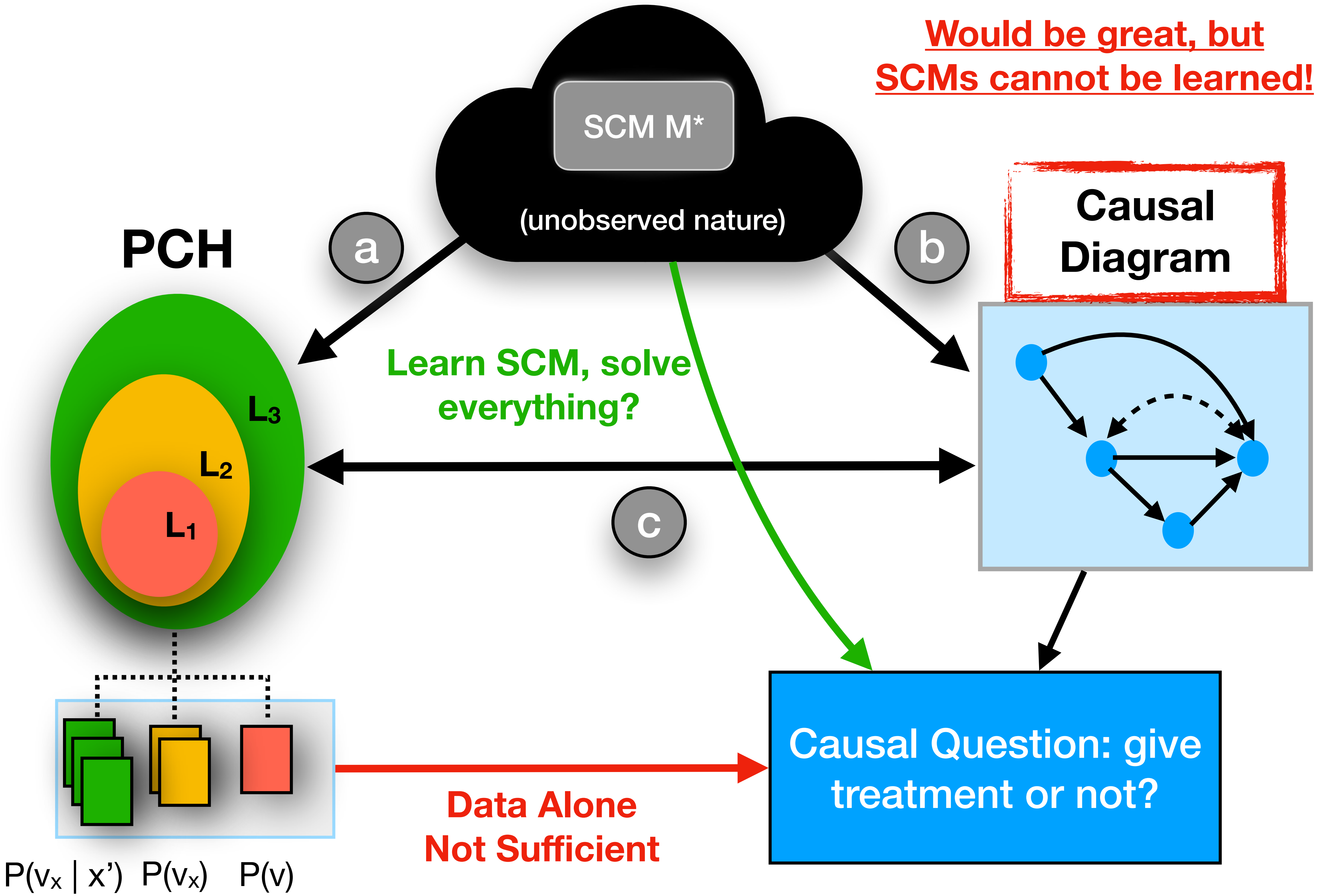
e.g. $y = \alpha + \beta X + U_Y$

Axiomatic Characterization:
(Galles-Pearl, 1998; Halpern, 1998).

Structural Causal Models



Are we aiming to learn the SCM?

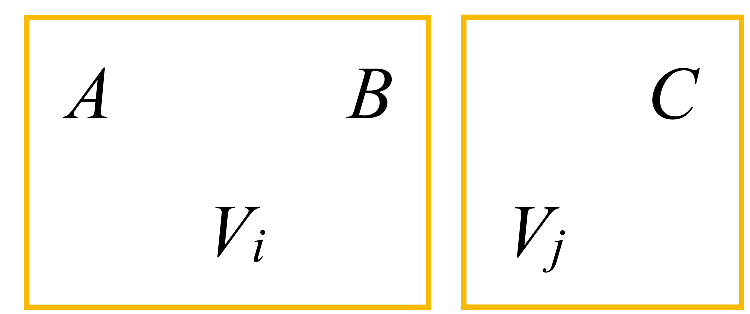


Property 0: SCM defines a causal diagram.

SCM → Causal Diagram

- Every SCM M induces a graphical model called causal diagram.
- Represented as a DAG where:
  - Each $V_i \in \boldsymbol{V}$ is a node,

$$V_i \leftarrow f_i\,(A,B,U)$$
$$V_j \leftarrow f_j\,(C,U)$$



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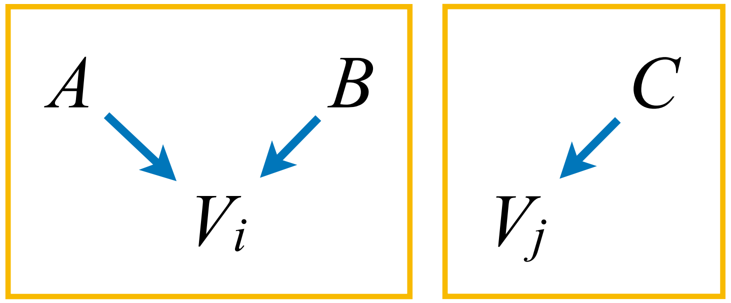

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  - There is $V_i \leftarrow\cdots\rightarrow V_j$ whenever $U_i \cap U_j \neq \varnothing$.

$$V_i \leftarrow f_i\,(A,B,U)$$
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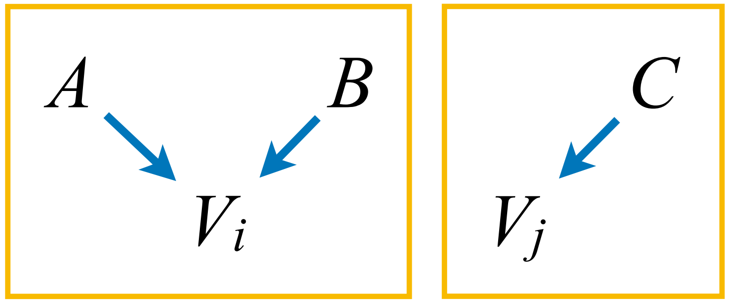

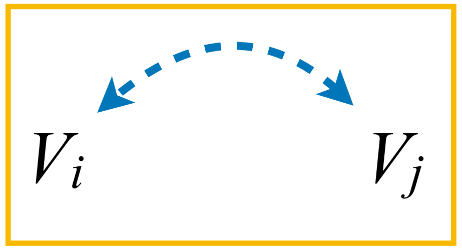

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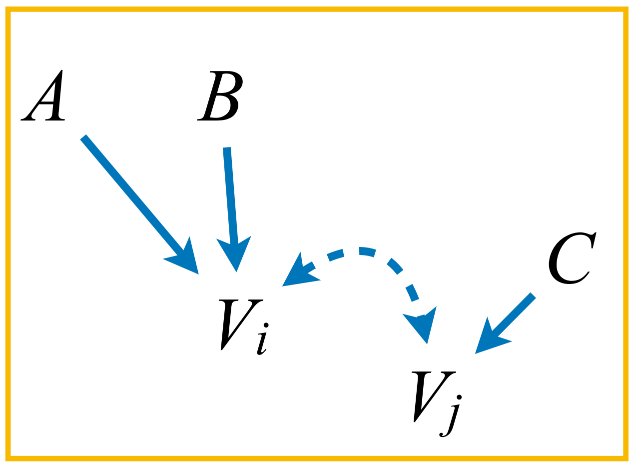

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Are we learning the SCM?

SCM Knowledge

very detailed knowledge

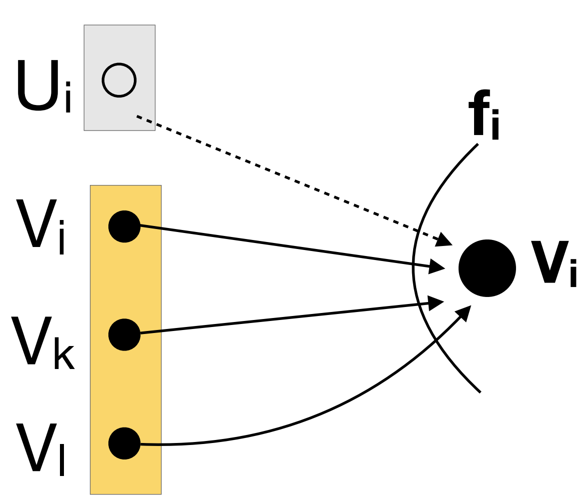
$$V_i \leftarrow \underbrace{\boxed{f_i}}(\mathrm{pa}(V_i), U_i)$$

exact functional form of how nature works

e.g., $V_i \leftarrow V_j^{2/3} \sin V_k + V_\ell \log U_i$

Diagram Knowledge

knowledge about f_i **arguments only**



Common Approach: Non-Parametric

- with the non parametric approach, we are trying to learn conditionals $P(V_i \mid \mathrm{pa}(V_i))$,
- we put no restriction on the possible class of functions we fit, meaning that conditionals are fully determined by the data.

Property 1: SCM induces L_1-distribution (observational).

SCM induces distribution $P(\mathbf{v})$

- **Observational Distribution.** An SCM $M = <\boldsymbol{V}, \boldsymbol{U}, \mathscr{F}, P(\boldsymbol{u})>$ defines a joint probability distribution $P^M(\boldsymbol{V})$:

$$\boxed{P^M(V = v) = \sum_{u \mid V(u) = v} P(U = u)}$$

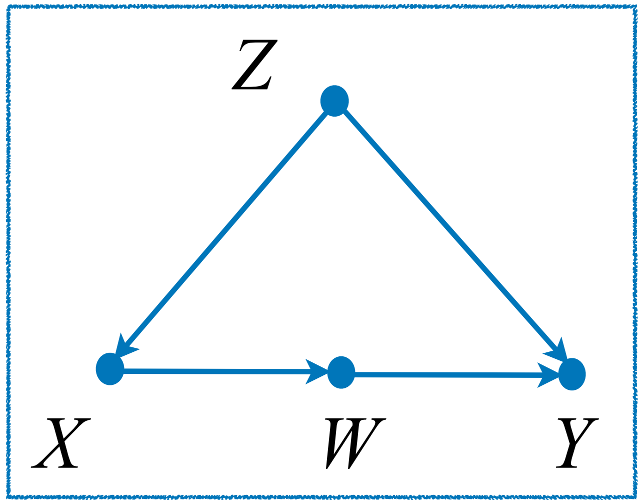
- $\mathscr{F}$ can be seen as a mapping from $\boldsymbol{U} \longrightarrow \boldsymbol{V}$

$$(u_1, u_2, \ldots, u_k) \longrightarrow \boxed{\mathscr{F}} \longrightarrow (v_1, v_2, \ldots, v_n)$$

SCM → Diagram → L1 Distribution: Markov Factorizations

- If no variable in $\boldsymbol{U}$ is a parent of more than one variable in $\boldsymbol{V}$ the model is called Markovian. Then:

Example



$$P(\mathbf{v}) = \prod_{V_i \in \mathbf{V}} P\left(v_i \mid pa_i\right)$$

known as Bayesian Factorization.

$P(v) =$
$P(z) \times$
$P(x \mid z) \times$
$P(w \mid x) \times$
$P(y \mid w, z)$

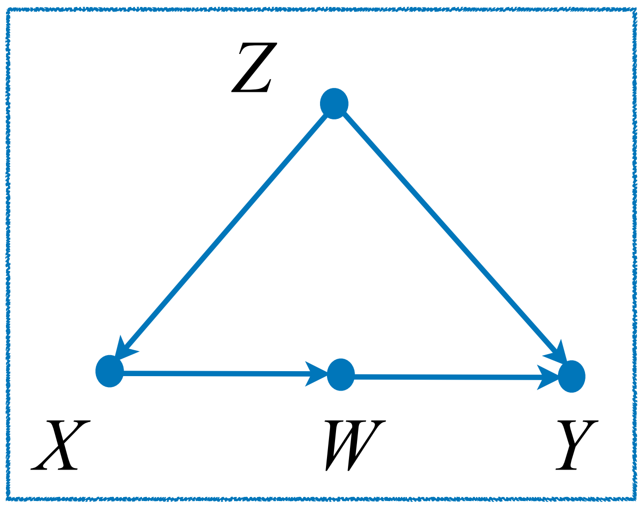
Graph implies conditional indepences.

e.g. $W \perp\!\!\!\perp Z \mid X$

Property 2: SCM induces L_2-distributions (interventional).

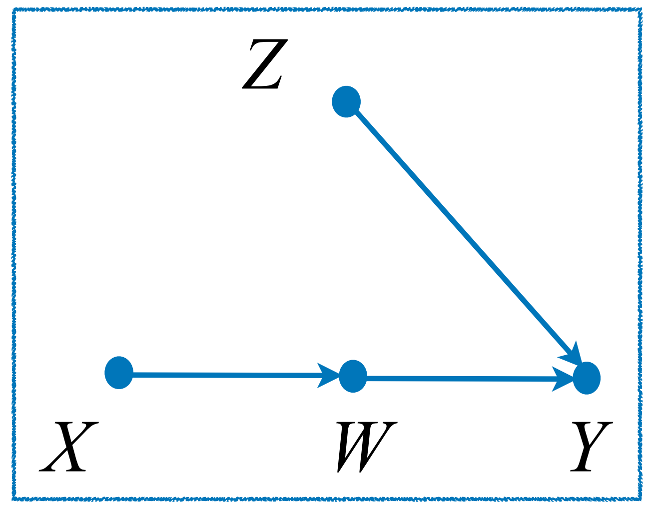
Defining Causal Effects

Real world



change

Alternative world



Z : cofounders

X : treatment

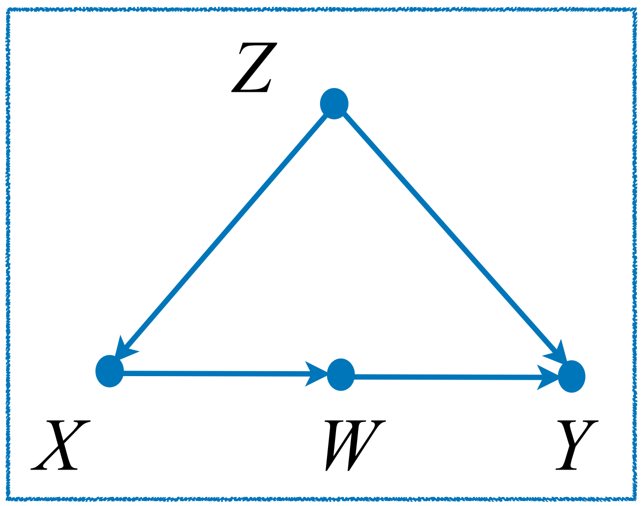
W : mediator

Y : outcome

$$M = \begin{cases} Z \leftarrow f_Z(u_z) \\ X \leftarrow f_X(z, u_x) \\ W \leftarrow f_W(x, u_w) \\ Y \leftarrow f_Y(w, z, u_y) \end{cases} \xrightarrow[do(X=x)]{} M_x = \begin{cases} Z \leftarrow f_Z(u_z) \\ \cancel{X \leftarrow f_X(z, u_x)} \quad X = x \\ W \leftarrow f_W(x, u_w) \\ Y \leftarrow f_Y(w, z, u_y) \end{cases}$$

Defining Causal Effects

Real world



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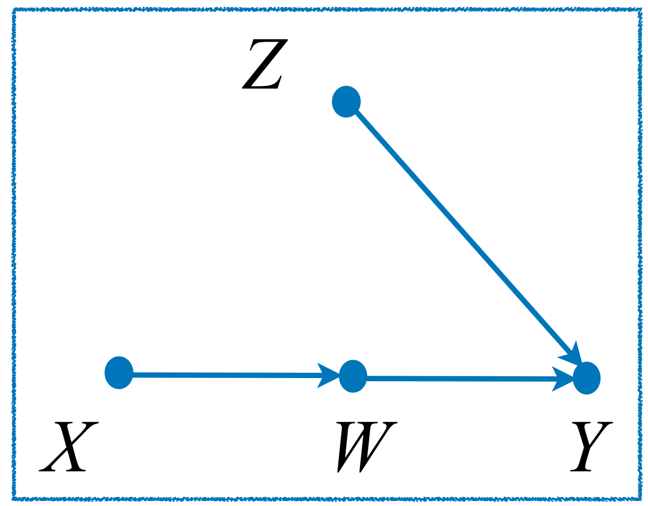

Z : cofounders

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Y : outcome

$P(v) =$

$P(z) \times$
$P(x \mid z) \times$
$P(w \mid x) \times$
$P(y \mid w, z)$

$\xrightarrow{}$
$do(X{=}x)$

$P_x(v) =$

$P(z) \times$
$\cancel{P(x \mid z) \times}$ equal to 1 in M_x
$P(w \mid x) \times$
$P(y \mid w, z)$

Property 3: SCM induces L_3-distributions (counterfactual).

Counterfactuals' Semantics

- Definition (Potential Outcome): Let $X, Y \subseteq V$.
  The potential outcome of Y to action $do(X = x)$, denoted by $Y_x(u)$, is the value of Y obtained for unit $U = u$ in the model M_x, where the equations of X are replaced with x.

$$
\begin{array}{l}
X \leftarrow f_x(u_x) \\
W \leftarrow f_w(X, u_w) \\
\vdots \\
Y \leftarrow f_y(X, W, u_y)
\end{array}
\xrightarrow{\text{Intervention}}
\left.
\begin{array}{l}
X \leftarrow x \\
W_x \leftarrow f_w(X, u_w) \\
\vdots \\
Y_x \leftarrow f_y(X, W, u_y)
\end{array}
\right\}
$$

$Y_x(u)$ is the value obtained by plugging in $U = u$ in this set of equations

Sometimes called a Counterfactual: *the value Y would have obtained, had X been x for the unit $U = u$*

Observational & Counterfactual Distributions

- For counterfactual quantities, their distribution can be defined via the SCM $\mathscr{M} = \langle V, U, \mathscr{F}, P(u) \rangle$, which induces a family of joint distributions over counterfactual events $Y_x = y, \ldots, Z_w = z$ for any $Y, Z, \ldots, X, W \subseteq V$:

$$P^{\mathscr{M}}(Y_x = y, \ldots, Z_w = z) = \sum_u \mathbf{1}\Big(Y_x(u) = y, \ldots, Z_w(u) = z\Big) P(u) \, .$$

- A special case of this, when the subscripts $x, \ldots, z$ are empty, gives the so-called observational distribution. In that case, we simply consider a set of variables $Y \subseteq V$ and the observational distribution is defined by:

$$P^{\mathscr{M}}(Y = y) = \sum_u \mathbf{1}\Big(Y(u) = y\Big) P(u) \, .$$

Big picture - Pearl's Causal Hierarchy

Level (Symbol)	Typical Activity	Typical Question	Examples
1 👀 Associational $P(y \mid x)$	Seeing (Associations)	What is? How would seeing X change my belief in Y?	Is low blood pressure associated with sepsis?
2 💪 Interventional $P(y \mid do(x), c)$	Doing (Treatment Effects, Interventions)	What if? What if I do X?	Do statins prevent heart attacks?
3 🧠 Counterfactual $P(y_x \mid x', y')$	Imagining, Retrospection (Treatment Harm)	Why? What if I had acted differently?	Had the patient not undergone surgery, would they be alive?

Big picture - Pearl’s Causal Hierarchy

Causal Inference = Cross-layer inferences

Level (Symbol)		Examples
1 Associa... $P(y \mid$ ...		...is low blood pressure ...ssociated ...th sepsis?
Interven...		
Counterfactua... $P(y_x \mid x', y')$		

Task-type 2

Task-type 3

Data: Layer 1, 2
Inference: Layer 3

e.g., *“if we adopted a different treatment policy, which patients would avoid harm?”*

Part II:
Why is this useful in practice?

Benefit 1: Encoding assumptions *transparently*.

Conditional Ignorability

- Q: What is conditional ignorability?

Definition. *Let X be a treatment variable, and let Y denote the outcome variable. We say that the pair (X, Y) satisfies conditional ignorability w.r.t. Z if*

$$Y_x \perp\!\!\!\perp X \mid Z$$

most common assumption in causal inference -> allows us to "adjust for Z"

- Q: Why is conditional ignorability useful?

$$P(Y_x = y) = \sum_z P(Y_x = y \mid Z = z)P(Z = z) \quad \text{Law of Tot. Prob.}$$

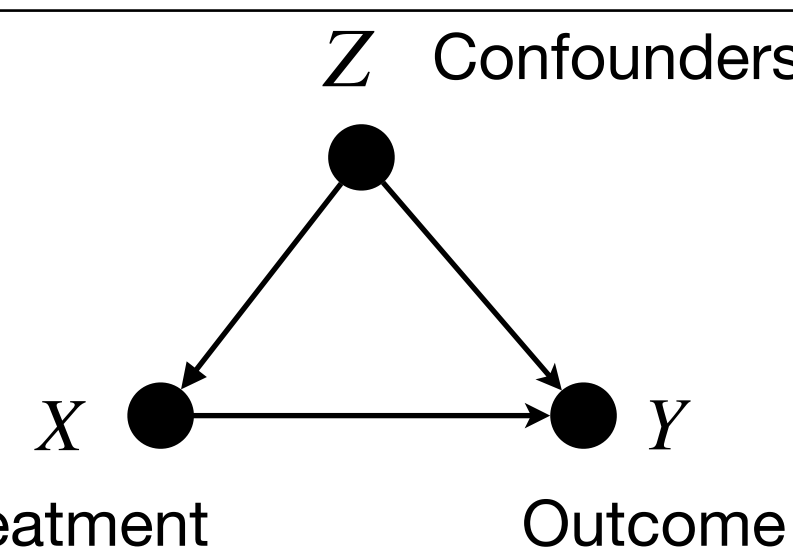
$$= \sum_z P(Y_x = y \mid Z = z, X = x)P(Z = z) \quad \text{Cond. Ign.}$$

$$= \sum_z P(Y = y \mid Z = z, X = x)P(Z = z) \quad \text{Consistency Axiom}$$

Conditional Ignorability: Pros & Cons

Benefits	Drawbacks?
(i) allows us to use adjustment	1 is ignorability easy to assess? i.e., do we have an algorithm?
(ii) simple to write down	2 can we express ignorability in human-understandable language?
(iii) does not require a full causal diagram	

many people believe:
ignorability $\Longleftrightarrow$ No confounding



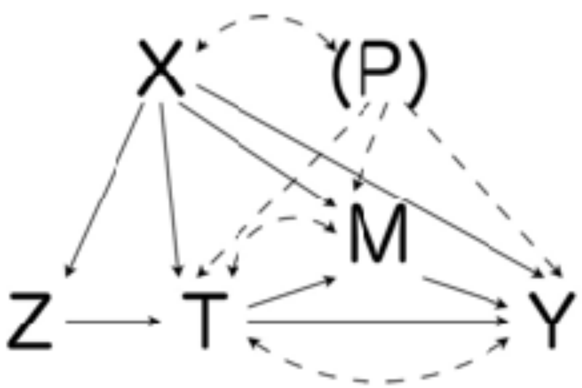
Causal: $Y_x \perp\!\!\!\perp X \mid Z$

how do we know whether this holds?

$\neq$

Statistical: $Y \perp\!\!\!\perp X \mid Z$

Ignorability Can Be Even More Elusive? An Example from Literature



(b)

Assumption 3 (Conditionally Ignorable Treatment Assignment)

$$\{Y_i(t,m), M_i(t'), T_i(z) : t, t', z \in \{0,1\}, m \in \mathcal{M}\} \perp\!\!\!\perp Z_i \mid X_i$$

Assumption 4 (Conditionally Ignorable Observed Mediator among Compliers)

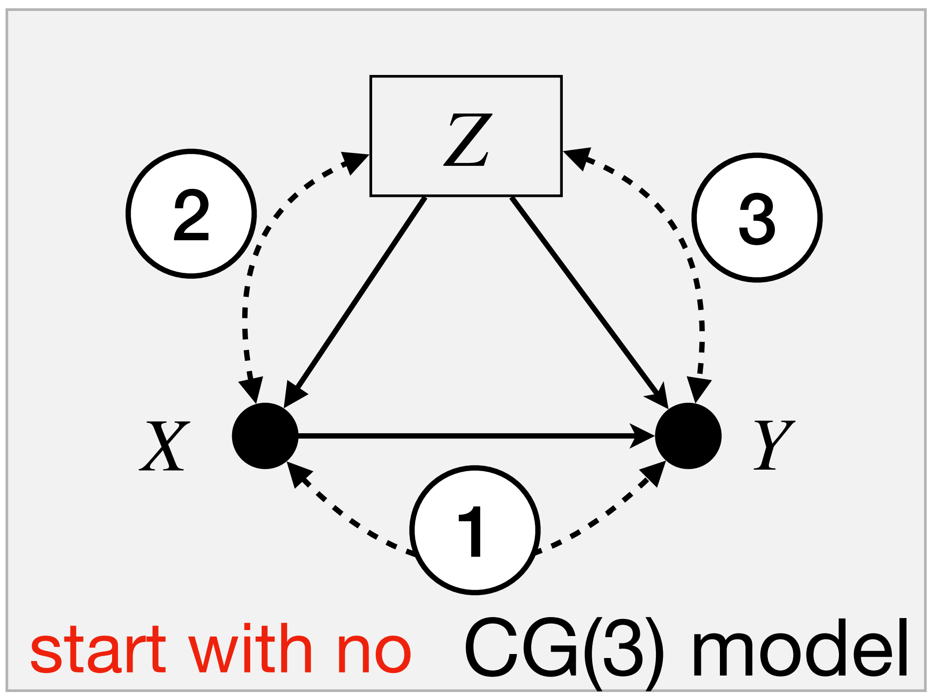
$$Y_i(t', m) \perp\!\!\!\perp M_i(t) \mid T_i = t, P_i = c, X_i,$$

for all $t, t' \in \{0,1\}$ *and* $m \in \mathcal{M}$.

The statement is hard to understand for $Y_x \perp\!\!\!\perp X \mid Z$ — and what about this?

Solution: Structural-Graphical Ignorability

Construct graph over three blocks



start with no assumptions

Elicit Causal Knowledge Stepwise

1 is there confounding X - Y?

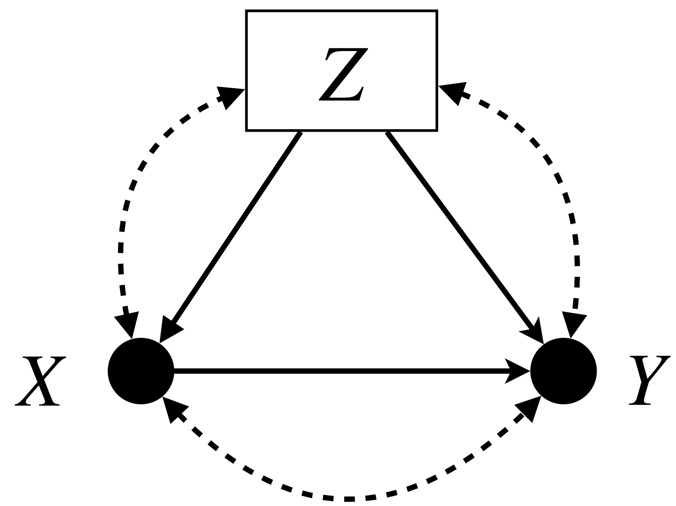
2 is there confounding X - Z?

3 is there confounding Y - Z?

Causal Assumptions encoded in edge removals!

The knowledge you elicit may be helpful for someone else's study.

Ignorability: Wrong Vehicle for Communication



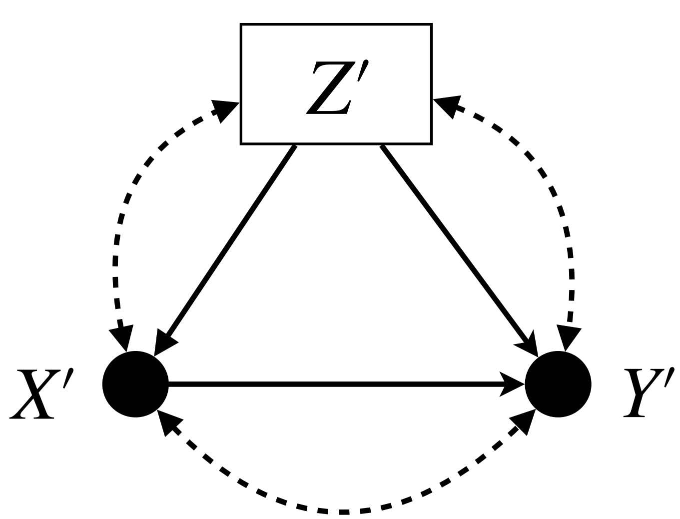
Team 1

Studies the causal effect of smoking (X) on systolic blood pressure (Y). Confounders include BMI, activity, cholesterol, demographics, and alcohol.

In Step 2, the team elicits important confounders between X and BMI — health awareness and SES.

In Step 3, the team elicits important confounders between Y and BMI — cardiometabolic genetic factors.

If you just write $Y_x \perp\!\!\!\perp X \mid Z$, noone else can learn from it.



Team 2

Studies the causal effect of smoking (X) on <u>BMI</u> (Y').

Causal knowledge from Team 1 ***transfers*** onto their study: HA and SES confound X and Y', while cardiometabolic factors confound Y' and Z.

Benefit 2: Leverage algorithmic machinery.

The Back-door Criterion

Definition (Back-door Criterion)

A set Z satisfies the back-door criterion (bdc) w.r.t. to a pair of variables X,Y in a causal diagram G if:

(i) no node in Z is a descendent of X; and

(ii) Z blocks every path between X and Y that contains an arrow into X.

The Back-door Adjustment

Theorem (Back-door Adjustment)

If a set Z satisfies the bdc w.r.t the pair X, Y, the effect of X on Y is identifiable and given by:

$$P(\mathbf{y} \mid do(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$$

Reduces a <u>Layer 2</u> object $P(y \mid do(x))$
to a <u>Layer 1</u> object $\sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$

Back to Sex-Stratification vs. Weight-Loss Stratification: Which table to consult?

$(\neg F)$	HbA1c low (Y)	HbA1c high $(\neg Y)$		Sucess Rate
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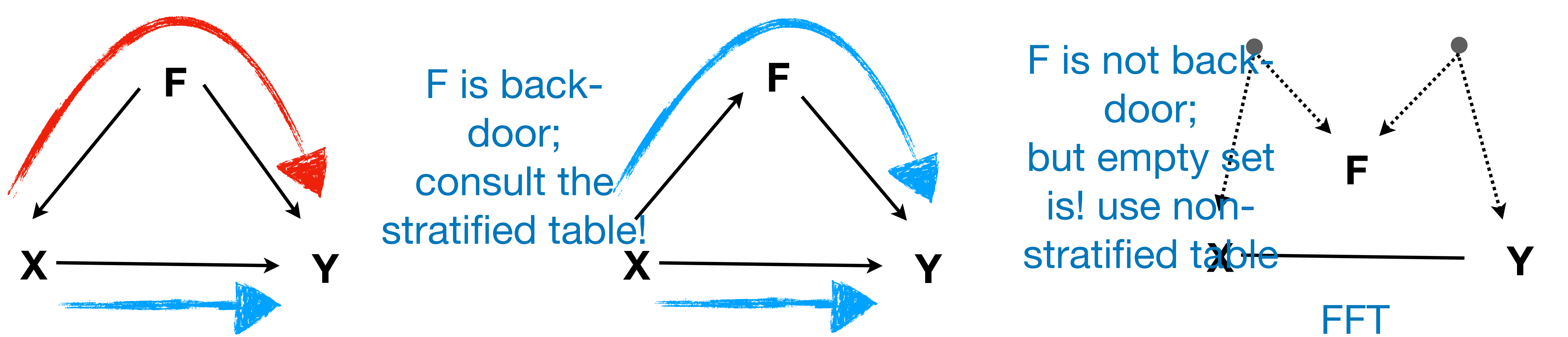
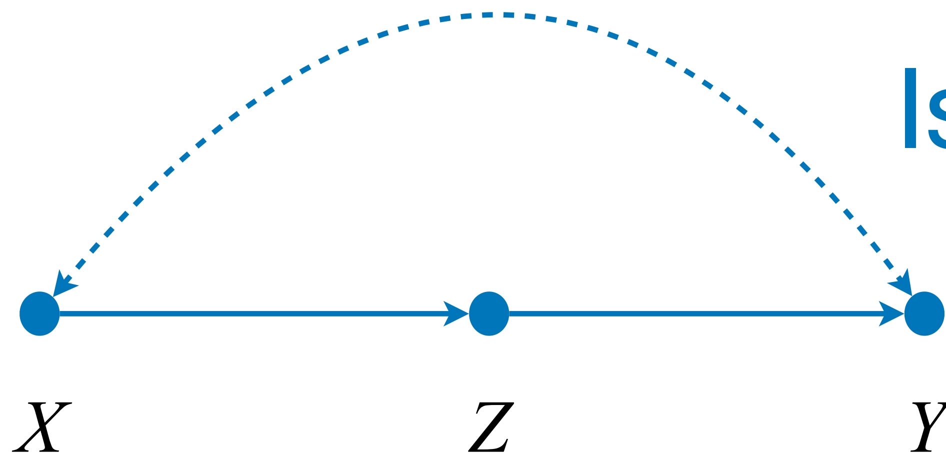
Sex-Stratified **Weight Loss-Stratified** **Another Story?**



Beyond the Back-Door

- Is the back-door criterion all that there is?



Is this model ID?

$$P(y \mid do(x)) = \sum_z P(z \mid x) \sum_{x'} P(y \mid z, x') P(x')$$

This is known as front-door identification.

Rules of Do-Calculus

Theorem. The following transformations are valid for any do-distribution induced by a SCM M:

Rule 1: Adding/removing Observations

$P(y|do(x),z,w)=P(y|do(x),w)$ if $(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}}}$

Rule 2: Action/observation exchange

$P(y|do(x),do(z),w)=P(y|do(x),z,w)$ if $(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}\underline{Z}}}$

Rule 3: Adding/removing Actions

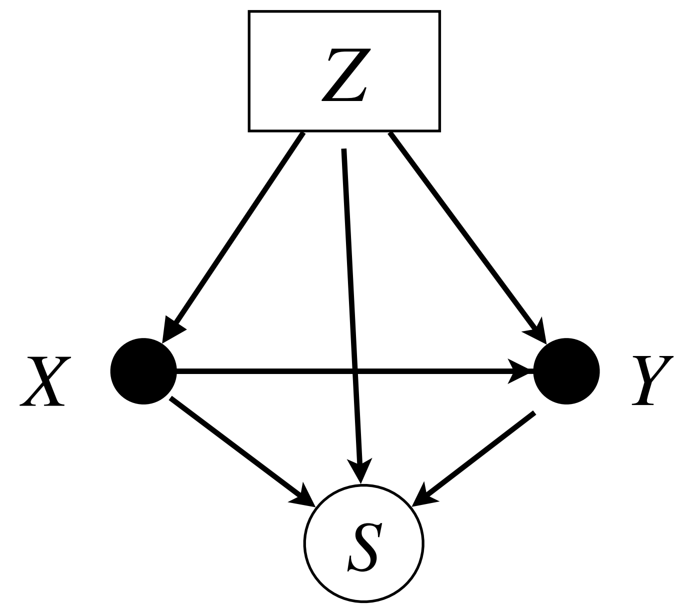
$P(y|do(x),do(z),w)=P(y|do(x),w)$ if $(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}\overline{Z(W)}}}$

where $Z(W) = Z \setminus \mathrm{an}(W)_{G_{\overline{X}}}$ is the subset of Z that are not ancestors of W in $G_{\overline{X}}$.

Benefit 3: Use DAGs to go beyond basic settings.

Selection Bias

- Suppose you study the effectiveness of a job-training program,
- X training, Y earnings, Z SES and motivation,



- What if you only have follow-up survey data?

Can we recover the effect?

  - (a) higher SES people more likely to respond
  - (b) training and SES increase response
  - (c) training, SES, and earnings increase response

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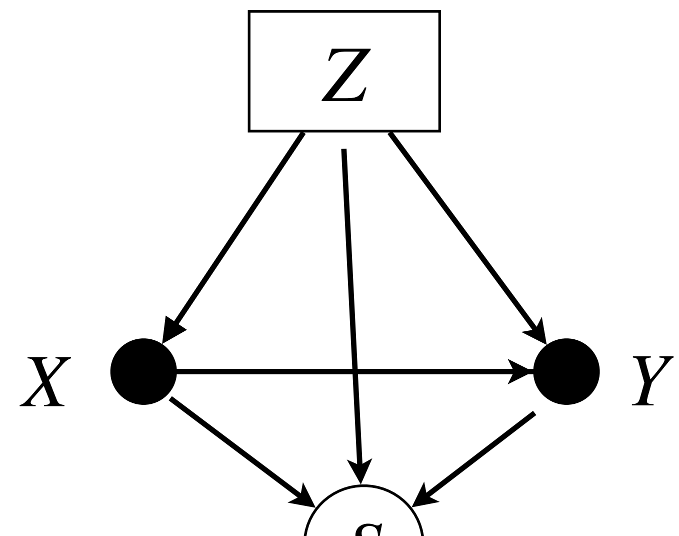


- Wh
  fro

Graphical models give powerful tools for studying selection bias.

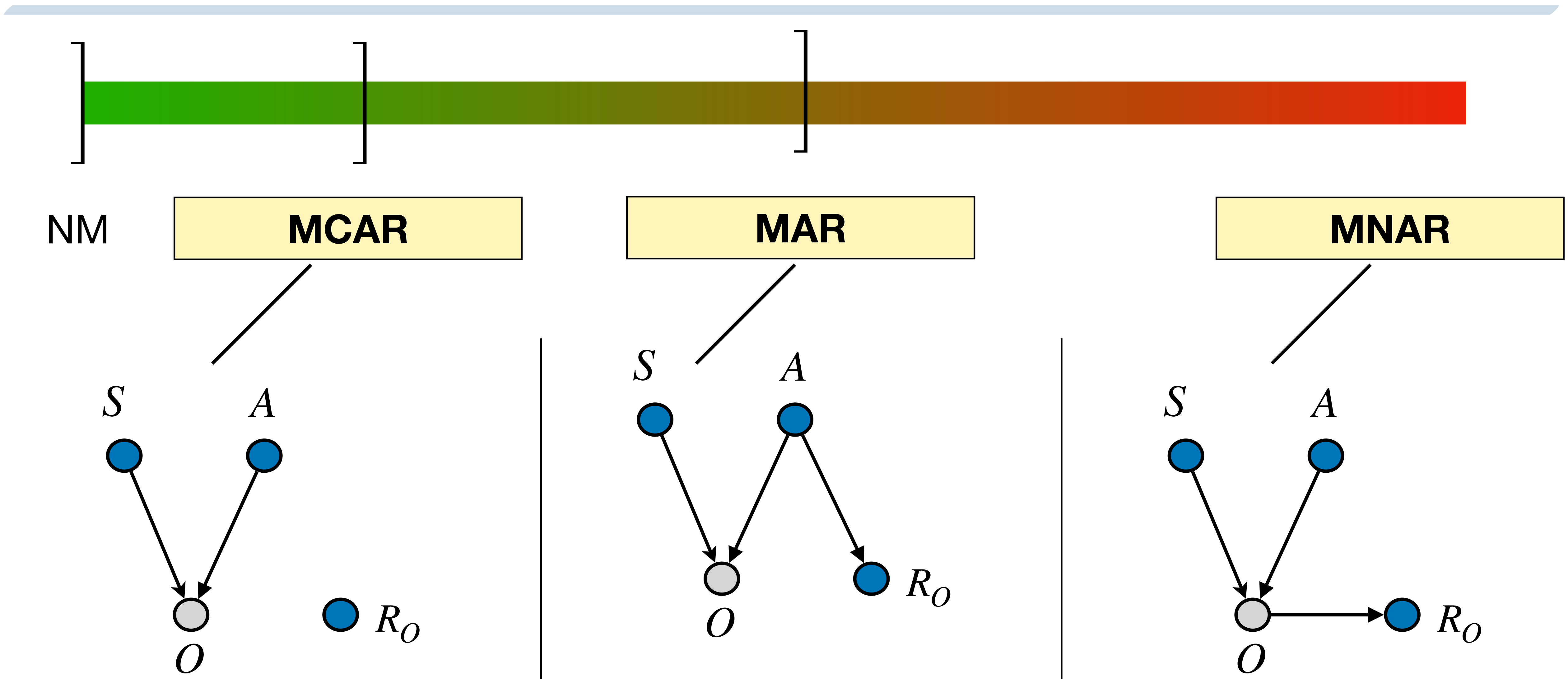
  - (
  - (b) training and SES increase response
  - (c) training, SES, and earnings increase response

Missing Data

- Suppose we are trying to analyze obesity in a group of students,
- We collect age, sex, and obesity,
- However, some values for obesity are not reported — i.e., the data is *missing*.
- How do we proceed with causal inference?

Age	Sex	Obesity
16	*F*	*Obese*
15	*F*	***m***
15	*M*	***m***
14	*F*	*Not Obese*
13	*M*	*Not Obese*
15	*M*	*Obese*
14	*F*	*Obese*

Missing Data

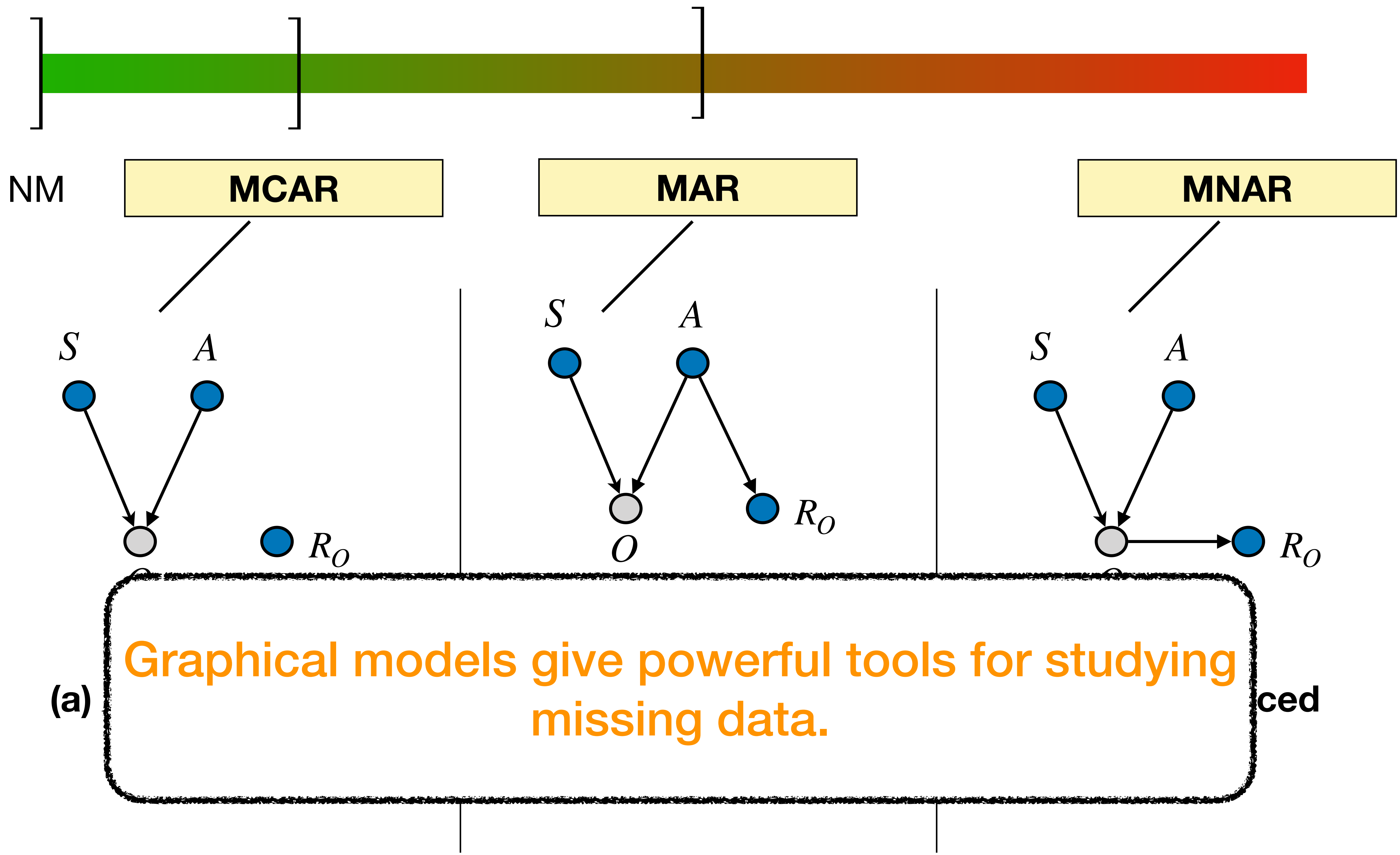


(a) R_O not influenced by any V

(b) R_O influenced by A

(c) R_O influenced by O

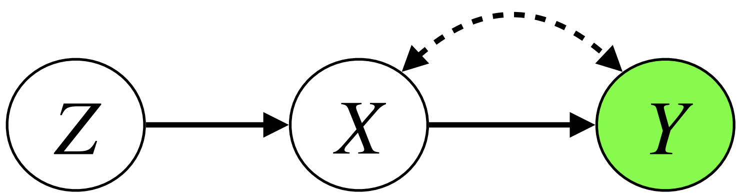
Missing Data



Graphical models give powerful tools for studying missing data.

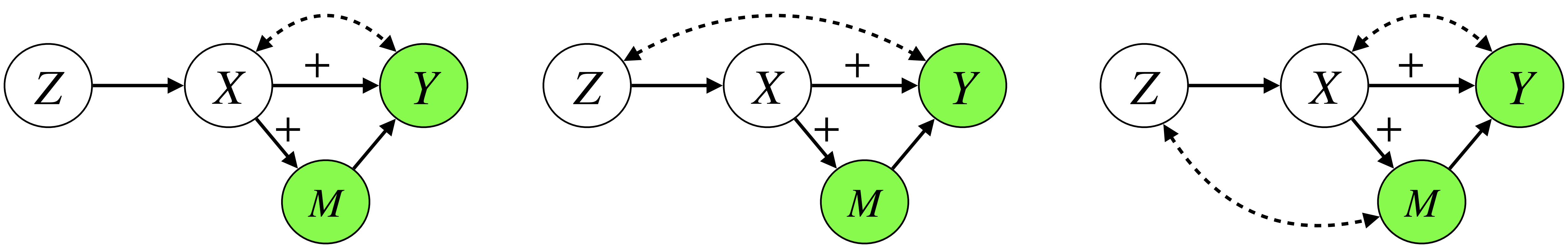
Shape-Constraints

- Shape-constraints may be a powerful tool for identifying causal effects,



LATE $:= E[Y_{x_1} - Y_{x_0} \mid X_{z_0} = 0, X_{z_1} = 1]$

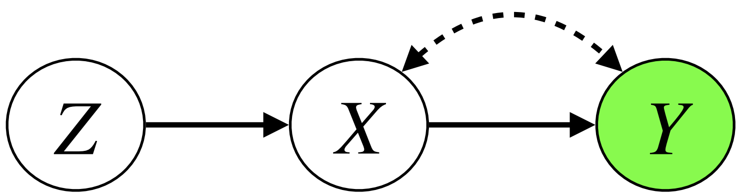
- What if we are in a more complicated setting?



(a) (b) (c)

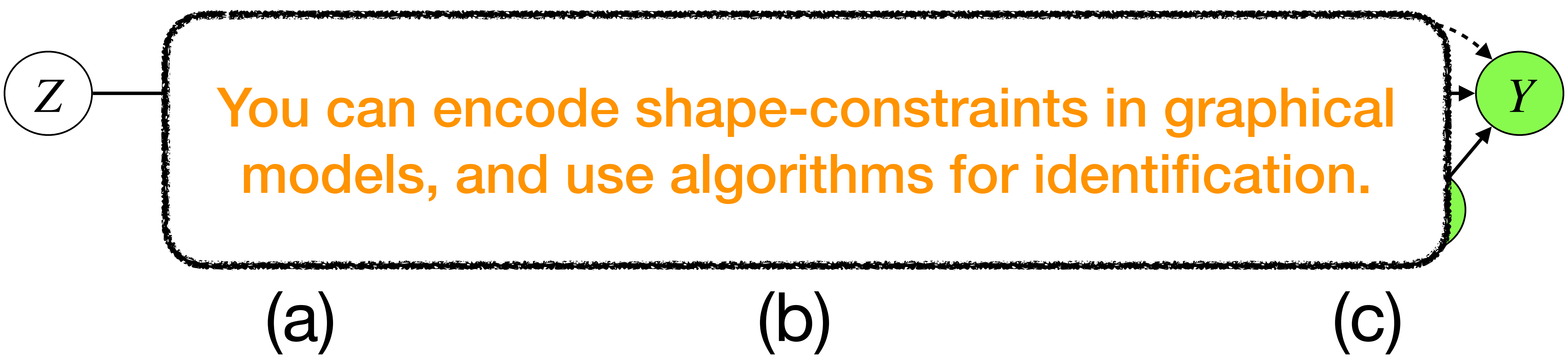
Shape-Constraints

- Shape-constraints may be a powerful tool for identifying causal effects,



$$\textbf{LATE} := E[Y_{x_1} - Y_{x_0} \mid X_{z_0} = 0, X_{z_1} = 1]$$

- What if we are in a more complicated setting?



You can encode shape-constraints in graphical models, and use algorithms for identification.